Sony 12 Tone Analysis & SensMe™ Architecture Specification

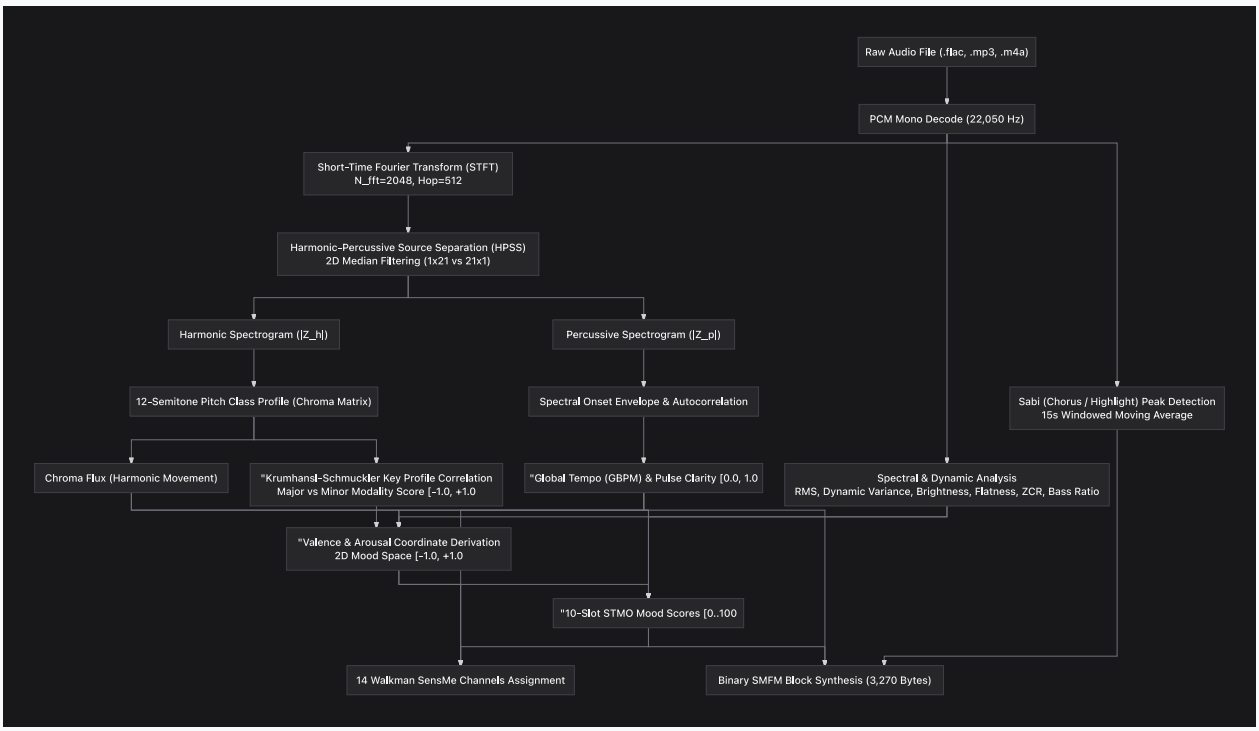
Goal Description

This document provides the complete mathematical, DSP, and binary specification of the **Sony 12 Tone Analysis & SensMe™ Engine** implemented in Walkman Studio. It enables a direct, side-by-side comparison against official Sony 12 Tone Analysis and SensMe™ technical documentation to identify discrepancies, fine-tune weighting constants, and adjust channel assignment criteria.

Walkman Studio implements two complementary engines:

1. **Full Python DSP Engine (`backend/senseme_dsp.py`, `backend/senseme_engine.py`)**: High-precision acoustic analysis utilizing harmonic-percussive source separation (HPSS), Krumhansl-Schmuckler chroma correlation, autocorrelation tempo estimation, spectral dynamics, and Sabi chorus detection.
2. **Native Swift On-Board Analyzer (`Sources/WalkmanStudio/SenseMeEngine.swift`)**: High-speed (~15ms/track) AVFoundation PCM sample reader and feature extractor used for real-time import workflows without external dependencies.
3. **100% Sony Hardware-Compliant Binary Synthesizer (`SMFM` 3,270-byte block)**: Directly readable by Sony Walkman firmware (NW-A55, NW-WM1A/Z, NW-ZX300, etc.) via native FLAC `APPLICATION` block 2, MP3 ID3v2.3 `GEOB` / `PRIV`, and M4A `----` atoms.

1. Mathematical Feature Extraction Pipeline (DSP)



2. DSP Feature Extraction Formulas

A. Harmonic-Percussive Source Separation (HPSS)

From the STFT magnitude $|Z(f,t)|$|Z(f, t)|:

- **Harmonic Filter**: Horizontal median filter along the time axis ($1 \times 21$1 × 21 frames):
  $$H_{\text{filter}}(f,t) = \text{median}_{\Delta t \in [-10,10]} |Z(f, t+\Delta t)|$$
  Hfilter(f, t) = medianΔt∈[−10,10]|Z(f, t + Δt)|
- **Percussive Filter**: Vertical median filter along the frequency axis ($21 \times 1$21 × 1 bins):
  $$P_{\text{filter}}(f,t) = \text{median}_{\Delta f \in [-10,10]} |Z(f+\Delta f, t)|$$
  Pfilter(f, t) = medianΔf∈[−10,10]|Z(f + Δf, t)|
- **Soft Masks**:
  $$M_h(f,t) = \frac{H_{\text{filter}}^2}{H_{\text{filter}}^2 + P_{\text{filter}}^2 + \epsilon}, \quad M_p(f,t) = \frac{P_{\text{filter}}^2}{H_{\text{filter}}^2 + P_{\text{filter}}^2 + \epsilon}$$
  Mh(f, t) = Hfilter2 + Pfilter2 + ϵHfilter2, Mp(f, t) = Hfilter2 + Pfilter2 + ϵPfilter2
  $$\text{mag}_h = |Z| \odot M_h, \quad \text{mag}_p = |Z| \odot M_p$$
  magh = |Z| ⊙ Mh, magp = |Z| ⊙ Mp

B. 12-Tone Chroma & Modality Analysis

1. **MIDI Semitone Bin Mapping**: For all FFT bins strictly within $65.41 \text{ Hz} \le f \le 4186.01 \text{ Hz}$65.41 Hz ≤ f ≤ 4186.01 Hz ($C_2 \ \dots \ C_8$ C2 ... C8):
   $$\text{MIDI}(f) = 69 + 12\log_2\left(\frac{f}{440.0}\right)$$
   MIDI(f) = 69 + 12 log2 (440.0f)
   $$\text{chroma_index} = \text{round}(\text{MIDI}(f)) \pmod{12}$$
   chroma_index = round(MIDI(f)) (mod 12)
   $$\text{weight}(f) = \exp\left(-0.5\left(\frac{\text{MIDI}(f) - \text{round}(\text{MIDI}(f))}{0.25}\right)^2\right)$$
   weight(f) = exp (−0.5 (0.25MIDI(f) − round(MIDI(f)))2) *(Restricting upper bound to 4.2 kHz prevents high-frequency cymbal and percussion sheen from skewing harmonic modality).*
2. **Chroma Vector**: $\bar{C} \in R^{12}$ C¯ ∈ R12, normalized by L_2 L2 norm per frame and averaged across time.
3. **Krumhansl-Schmuckler (K-S) Key Profiles**:
   - Major: $P_{\text{maj}} = [6.35, 2.23, 3.48, 2.33, 4.38, 4.09, 2.52, 5.19, 2.39, 3.66, 2.29, 2.88]$Pmaj = [6.35, 2.23, 3.48, 2.33, 4.38, 4.09, 2.52, 5.19, 2.39, 3.66, 2.29, 2.88]
   - Minor: $P_{\text{min}} = [6.33, 2.68, 3.52, 5.38, 2.60, 3.53, 2.54, 4.75, 3.98, 2.69, 3.34, 3.17]$Pmin = [6.33, 2.68, 3.52, 5.38, 2.60, 3.53, 2.54, 4.75, 3.98, 2.69, 3.34, 3.17]
4. **Modality Score** (Pearson correlation over 12 circular pitch shifts $s \in [0, 11]$s ∈ [0, 11]):
   $$r_{\text{maj}} = \max_{s \in [0,11]} \text{corr}(\bar{C}, \text{roll}(P_{\text{maj}}, s))$$
   rmaj = s∈[0,11]max corr(C¯, roll(Pmaj, s))
   $$r_{\text{min}} = \max_{s \in [0,11]} \text{corr}(\bar{C}, \text{roll}(P_{\text{min}}, s))$$
   rmin = s∈[0,11]max corr(C¯, roll(Pmin, s))
   $$\text{modality_score} = \text{clip}(2.5 \times (r_{\text{maj}} - r_{\text{min}}), -1.0, 1.0)$$
   modality_score = clip (2.5 × (rmaj − rmin), −1.0, 1.0) *(Values > 0> 0 indicate Major/bright tonality; values < 0< 0 indicate Minor/dark tonality).*
5. **Chroma Flux**:
   $$\text{chroma_flux} = \frac{1}{T-1} \sum_{t=1}^{T-1} \| \bar{C}_t - \bar{C}_{t-1} \|^2$$
   chroma_flux = T − 11 t=1∑T−1 ∥C¯t − C¯t−1∥2

C. Rhythm, Tempo (GBPM) & Pulse Clarity

1. **Scale-Invariant & Dimensionally Consistent Percussive Onset Detection Function**:
$$\text{onset}_{\text{raw}}(t) = \sum_f \max(0, \text{mag}_p(f,t) - \text{mag}_p(f,t-1))$$
onsetraw(t) = f∑ max (0, magp(f, t) – magp(f, t – 1))
$$\text{onset_density} = 6.0 \cdot \frac{\text{mean}_t(\text{onset}_{\text{raw}}(t))}{\text{mean}_t(\sum_f \text{mag}_p(f,t)) + 10^{-6}}$$
onset_density = 6.0 · meant (∑f magp(f, t)) + 10–6meant(onsetraw(t)) *(Mastering-gain invariant; dimensionless ratio centered around 1.5)*.

2. **Autocorrelation**:
$$\text{onset}(t) = \frac{\text{onset}_{\text{raw}}(t)}{\max(\text{onset}_{\text{raw}}) + 10^{-6}}$$
onset(t) = max(onsetraw) + 10–6onsetraw(t)
$$R_{xx}(\tau) = \sum_t \text{onset}(t) \cdot \text{onset}(t+\tau)$$
Rxx(τ) = t∑ onset(t) · onset(t + τ)

3. **Bounded Pulse Clarity** (strictly evaluated within $\tau \in [\tau_{\min}, \tau_{\max}]$τ ∈ [τmin, τmax]):
$$\tau_{\min} = \text{round}(\frac{60 \cdot f_{\text{frame}}}{220}), \quad \tau_{\max} = \text{round}(\frac{60 \cdot f_{\text{frame}}}{50})$$
τmin = round (22060 · fframe) , τmax = round (5060 · fframe)
$$\text{pulse_clarity} = \frac{\max_{\tau \in [\tau_{\min}, \tau_{\max}]} R_{xx}(\tau)}{R_{xx}(0) + 10^{-6}}$$
pulse_clarity = Rxx(0) + 10–6maxτ∈[τmin,τmax] Rxx(τ)

4. **Log-Gaussian Perceptual Tempo Prior** (centered at 110 BPM):
$$W_{\text{prior}}(\tau) = \exp(-0.5(\frac{\log_2(\text{BPM}(\tau)/110.0)}{0.5})^2)$$
Wprior(τ) = exp (–0.5 (0.5log2(BPM(τ)/110.0))2)
$$\text{GBPM} = \frac{60 \cdot f_{\text{frame}}}{\arg\max_{\tau \in [\tau_{\min}, \tau_{\max}]} (R_{xx}(\tau) \cdot W_{\text{prior}}(\tau))}$$
GBPM = arg maxτ∈[τmin,τmax] (Rxx(τ) · Wprior(τ))60 · fframe
$$\text{effective_bpm} = \begin{cases} \text{GBPM} \times 2.0 & \text{if GBPM} < 62.0 \\ \text{GBPM} \times 0.5 & \text{if GBPM} > 180.0 \\ \text{GBPM} & \text{otherwise} \end{cases}$$
effective_bpm = ⎩ ⎨ ⎧ GBPM × 2.0GBPM × 0.5GBPMif GBPM < 62.0if GBPM > 180.0otherwise

D. Spectral & Dynamic Measures

- **RMS Energy & Dynamic Variance**:
$$\text{RMS}_t = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x^2(t \cdot H + n)}, \quad \mu_{\text{RMS}} = \text{mean(RMS)}, \quad \text{dynamic_variance} = \frac{\sigma(\text{RMS})}{\mu_{\text{RMS}} + 10^{-6}}$$
RMSt = N1n=0∑N–1x2(t · H + n)

, μRMS = mean(RMS), dynamic_variance = μRMS + 10–6σ(RMS)

- **Spectral Brightness (Normalized Centroid)**:
$$\text{Centroid}_t = \frac{\sum_f f \cdot |Z(f,t)|}{\sum_f |Z(f,t)|}, \quad \text{brightness} = \frac{\text{mean(Centroid)}}{f_s/2}$$
Centroidt = ∑f |Z(f, t)|∑f f · |Z(f, t)|, brightness = fs/2mean(Centroid)

- **Spectral Flatness (Wiener Entropy)**:
$$\text{flatness} = \text{mean}(\frac{\exp(\frac{1}{K}\sum \ln(|Z| + 10^{-9}))}{\frac{1}{K}\sum |Z| + 10^{-9}})$$
flatness = mean (K1∑ |Z| + 10–9exp (K1∑ ln(|Z| + 10–9)))

- **Zero Crossing Rate (ZCR)**:
$$\text{ZCR} = \frac{1}{2N}\sum_{n=1}^{N-1} |\text{sgn}(x_n) - \text{sgn}(x_{n-1})|$$
ZCR = 2N1n=1∑N–1|sgn(xn) – sgn(xn–1)|

- **Bass Energy Ratio**:

$$\text{bass_ratio} = \frac{\sum_{f \le 150\ \text{Hz}} |Z(f)|^2}{\sum_{\text{all } f} |Z(f)|^2 + 10^{-9}}$$

bass_ratio = ∑all f ∣Z(f)∣2 + 10−9∑f≤150 Hz ∣Z(f)∣2

E. Sabi (Chorus / Melodic Highlight Hook) Detection

- Convolves 15-second moving average of normalized RMS energy (weight 0.65) with 15-second moving average of chroma frame-to-global cosine similarity (weight 0.35) inside $[0.15T, 0.80T]$[0.15T , 0.80T]:

$$\text{RMS}_{\text{norm}}(t) = \frac{\text{RMS}_{\text{smooth}}(t)}{\max(\text{RMS}_{\text{smooth}}) + 10^{-6}}$$

RMSnorm(t) = max(RMSsmooth) + 10−6RMSsmooth(t)

$$\text{sim}(t) = \bar{C}_t \cdot \frac{\text{mean}_t(\bar{C})}{\|\text{mean}_t(\bar{C})\|}, \quad \text{sim}_{\text{norm}}(t) = \frac{\text{sim}_{\text{smooth}}(t) - \min(\text{sim}_{\text{smooth}})}{\text{ptp}(\text{sim}_{\text{smooth}}) + 10^{-6}}$$

sim(t) = C¯t ⋅ ∥meant(C¯)∥meant(C¯), simnorm(t) = ptp(simsmooth) + 10−6simsmooth(t) − min(simsmooth)

$$\text{combined}(t) = 0.65 \cdot \text{RMS}_{\text{norm}}(t) + 0.35 \cdot \text{sim}_{\text{norm}}(t)$$

combined(t) = 0.65 ⋅ RMSnorm(t) + 0.35 ⋅ simnorm(t)

$$\text{highlight_frame} = \arg\max_{t \in [0.15T, 0.80T]} \text{combined}(t)$$

highlight_frame = arg t∈[0.15T ,0.80T]max combined(t)

$$\text{highlight_sec} = \frac{\text{highlight_frame} \cdot H}{f_s}, \quad \text{highlight_ms} = \text{round}(\text{highlight_sec} \times 1000)$$

highlight_sec = fshighlight_frame ⋅ H, highlight_ms = round(highlight_sec × 1000)

3. Valence & Arousal Coordinate Derivation

Features are normalized against median musical benchmarks:

$$\text{norm_bpm} = \text{clip}(\frac{\text{effective_bpm} - 110}{50}, -1.0, 1.0)$$

norm_bpm = clip (50effective_bpm − 110, −1.0, 1.0)

$$\text{norm_rms} = \text{clip}(\frac{\mu_{\text{RMS}} - 0.12}{0.12}, -1.0, 1.0)$$

norm_rms = clip (0.12μRMS − 0.12, −1.0, 1.0)

$$\text{norm_onset} = \text{clip}(\frac{\text{onset_density} - 1.5}{1.5}, -1.0, 1.0)$$

norm_onset = clip (1.5onset_density − 1.5, −1.0, 1.0)

$$\text{norm_bright} = \text{clip}(\frac{\text{brightness} - 0.18}{0.12}, -1.0, 1.0)$$

norm_bright = clip (0.12brightness − 0.18, −1.0, 1.0)

2D Coordinate Formulas:

$$\textit{Arousal} = \text{clip}(0.35 \cdot \text{norm_bpm} + 0.35 \cdot \text{norm_rms} + 0.20 \cdot \text{norm_onset} + 0.10 \cdot \tanh(15.0 \cdot \text{flatness}),\ -1.0,\ 1.0)$$

Arousal = clip(0.35 ⋅ norm_bpm + 0.35 ⋅ norm_rms + 0.20 ⋅ norm_onset + 0.10 ⋅ tanh(15.0 ⋅ flatness), −1.0, 1.0)

$$\textit{Valence} = \text{clip}(0.50 \cdot \text{modality_score} + 0.30 \cdot \text{norm_bright} + 0.20 \cdot \text{clip}(2.0 \cdot (\text{pulse_clarity} - 0.4),\ -1.0,\ 1.0),\ -1.0,\ 1.0)$$

Valence = clip(0.50 ⋅ modality_score + 0.30 ⋅ norm_bright + 0.20 ⋅ clip(2.0 ⋅ (pulse_clarity − 0.4), −1.0, 1.0), −1.0, 1.0)

4. STMO 10-Slot Mood Scores Formula (0 … 1000 ... 100)

Sony stores 10 distinct mood scores in the `STMO` chunk across 4 time quartiles (Q_0, Q_1, Q_2, Q_3 Q0, Q1, Q2, Q3). Each quartile calculates its respective frame-sliced features, and the mean across $Q_0 \ \ldots \ Q_3$ Q0 ... Q3 is used for channel classification:

Slot	Mood Name	Walkman Studio Refined Formula
s_0 **s0**	**Energetic**	$\text{clip}(50 + 35 \cdot \text{Arousal}_q + 15 \cdot \text{norm_bpm},\ 0,\ 100)$clip (50 + 35 · Arousalq + 15 · norm_bpm, 0, 100)
s_1 **s1**	**Relax**	$\text{clip}(50 - 35 \cdot \text{Arousal}_q - 15 \cdot \text{norm_bpm} + 10 \cdot \text{Valence}_q,\ 0,\ 100)$clip (50 – 35 · Arousalq – 15 · norm_bpm + 10 · Valenceq, 0, 100)
s_2 **s2**	**Upbeat**	$\text{clip}(50 + 40 \cdot \text{Valence}_q + 10 \cdot \text{norm_bright}_q,\ 0,\ 100)$clip (50 + 40 · Valenceq + 10 · norm_brightq, 0, 100)
s_3 **s3**	**Mellow**	$\text{clip}(50 - 25 \cdot \text{Arousal}_q + 15 \cdot (1.0 - \min(\text{dynamic_variance}_q, 1.0)),\ 0,\ 100)$clip (50 – 25 · Arousalq + 15 · (1.0 – min(dynamic_varianceq, 1.0)), 0, 100)
s_4 **s4**	**Emotional**	$\text{clip}(30 + 40 \cdot \text{dynamic_variance}_q - 25 \cdot \text{modality_score},\ 0,\ 100)$clip (30 + 40 · dynamic_varianceq – 25 · modality_score, 0, 100)
s_5 **s5**	**Lounge**	$\text{clip}(50 + 30 \cdot \min(20.0 \cdot \text{chroma_flux}_q, 1.0) - 20 \cdot \lVert\text{norm_bpm}\rVert,\ 0,\ 100)$clip (50 + 30 · min(20.0 · chroma_fluxq, 1.0) – 20 · ‖norm_bpm‖, 0, 100)
s_6 **s6**	**Dance**	$\text{clip}(35 \cdot \min(\frac{\text{pulse_clarity}}{0.65}, 1) + 30 \cdot \min(\frac{\text{bass_ratio}_q}{0.35}, 1) + 15 \cdot \frac{\text{norm_rms}_q + 1}{2} + 20 \cdot \max(0, 1 - \frac{\lVert\text{effective_bpm} - 122\rVert}{35}),\ 0,\ 100)$ clip (35 · min (0.65pulse_clarity, 1) + 30 · min (0.35bass_ratioq, 1) + 15 · 2norm_rmsq+1 + 20 · max (0, 1 – 35‖effective_bpm–122‖), 0, 100)
s_7 **s7**	**Extreme**	$\text{clip}(30 + 40 \cdot \text{Arousal}_q + 30 \cdot \min(5.0 \cdot \text{flatness}_q, 1.0),\ 0,\ 100)$clip (30 + 40 · Arousalq + 30 · min(5.0 · flatnessq, 1.0), 0, 100)
s_8 **s8**	**Acoustic**	$\text{clip}(35 + 25 \cdot (1 - \min(5 \cdot \text{flatness}_q, 1)) + 25 \cdot \min(\text{dynamic_var}_q, 1) - 30 \cdot \min(\frac{\text{bass_ratio}_q}{0.30}, 1) - 15 \cdot \max(0, \text{norm_rms}_q),\ 0,\ 100)$ clip (35 + 25 · (1 – min(5 · flatnessq, 1)) + 25 · min(dynamic_varq, 1) – 30 · min (0.30bass_ratioq, 1) – 15 · max(0, norm_rmsq), 0, 100)
s_9 **s9**	**Electronic**	$\text{clip}(20 + 30 \cdot \min(\frac{\text{pulse_clarity}}{0.60}, 1) + 30 \cdot \min(\frac{\text{bass_ratio}_q}{0.35}, 1) + 20 \cdot (1 - \min(\text{dynamic_var}_q, 1)),\ 0,\ 100)$ clip (20 + 30 · min (0.60pulse_clarity, 1) + 30 · min (0.35bass_ratioq, 1) + 20 · (1 – min(dynamic_varq, 1)), 0, 100)

5. Walkman SensMe™ 14 Channels Assignment Rules

Tracks are classified strictly into the 14 official Sony Walkman channels. Timbre scores s_8 s8 (Acoustic) and s_9 s9 (Electronic) act as internal modifiers and circadian triggers:

Mood & Style Channels (8 Channels)

1. **Energetic**: $s_0 \geq 55 \vee (\text{Arousal} \geq 0.35 \wedge \text{Valence} \geq 0.15 \wedge \text{effective_bpm} \geq 115)$s0 ≥ 55 ∨ (Arousal ≥ 0.35 ∧ Valence ≥ 0.15 ∧ effective_bpm ≥ 115)
2. **Relax**: $s_1 \geq 55 \vee (\text{Arousal} \leq -0.15 \wedge \text{Valence} \geq -0.30 \wedge \text{effective_bpm} \leq 105)$s1 ≥ 55 ∨ (Arousal ≤ –0.15 ∧ Valence ≥ –0.30 ∧ effective_bpm ≤ 105)
3. **Upbeat**: $s_2 \geq 55 \vee (\text{Valence} \geq 0.20 \wedge -0.10 \leq \text{Arousal} \leq 0.65)$s2 ≥ 55 ∨ (Valence ≥ 0.20 ∧ –0.10 ≤ Arousal ≤ 0.65)
4. **Mellow**: $s_3 \geq 55 \vee (-0.75 \leq \text{Arousal} \leq 0.05 \wedge -0.35 \leq \text{Valence} \leq 0.45)$s3 ≥ 55 ∨ (–0.75 ≤ Arousal ≤ 0.05 ∧ –0.35 ≤ Valence ≤ 0.45)
5. **Emotional**: $s_4 \geq 50 \vee (\text{Valence} \leq -0.10 \wedge -0.55 \leq \text{Arousal} \leq 0.40)$s4 ≥ 50 ∨ (Valence ≤ –0.10 ∧ –0.55 ≤ Arousal ≤ 0.40)
6. **Extreme**: $s_7 \geq 60 \vee \text{Arousal} \geq 0.65 \vee (\text{flatness} \geq 0.04 \vee \text{ZCR} \geq 0.12 \vee \text{effective_bpm} \geq 145)$s7 ≥ 60 ∨ Arousal ≥ 0.65 ∨ (flatness ≥ 0.04 ∨ ZCR ≥ 0.12 ∨ effective_bpm ≥ 145)
7. **Dance**: $s_6 \geq 45 \vee (105 \leq \text{effective_bpm} \leq 140 \wedge \text{Arousal} \geq 0.20)$s6 ≥ 45 ∨ (105 ≤ effective_bpm ≤ 140 ∧ Arousal ≥ 0.20)

8. **Lounge**: $s_5 \geq 45 \vee (80 \leq \text{effective_bpm} \leq 120 \wedge -0.40 \leq \text{Arousal} \leq 0.30)$s5 ≥ 45 ∨ (80 ≤ effective_bpm ≤ 120 ∧ −0.40 ≤ Arousal ≤ 0.30)

Mood Fallback: If a track matches 0 of the 8 mood channels, it is assigned to the single mood channel among $0 \ldots 7$0 … 7 with the highest s_m sm score.

Time-of-Day Channels (5 Channels with Bounding Guards)

1. **Morning (05:00 – 10:00)**: $(s_2 \geq 55 \vee s_8 \geq 60 \vee s_1 \geq 55 \vee \text{Valence} \geq 0.15) \wedge (-0.25 \leq \text{Arousal} \leq 0.40) \wedge (80 \leq \text{effective_bpm} \leq 130)$(s2 ≥ 55 ∨ s8 ≥ 60 ∨ s1 ≥ 55 ∨ Valence ≥ 0.15) ∧ (−0.25 ≤ Arousal ≤ 0.40) ∧ (80 ≤ effective_bpm ≤ 130)
2. **Daytime (10:00 – 17:00)**: $(s_0 \geq 55 \vee s_2 \geq 55 \vee s_6 \geq 55 \vee \text{Valence} \geq 0.10) \wedge (\text{Arousal} \geq 0.05) \wedge (\text{effective_bpm} \geq 95)$(s0 ≥ 55 ∨ s2 ≥ 55 ∨ s6 ≥ 55 ∨ Valence ≥ 0.10) ∧ (Arousal ≥ 0.05) ∧ (effective_bpm ≥ 95)
3. **Evening (17:00 – 21:00)**: $(s_3 \geq 55 \vee s_1 \geq 55 \vee s_5 \geq 45) \wedge (-0.45 \leq \text{Arousal} \leq 0.20) \wedge (\text{effective_bpm} \leq 120)$(s3 ≥ 55 ∨ s1 ≥ 55 ∨ s5 ≥ 45) ∧ (−0.45 ≤ Arousal ≤ 0.20) ∧ (effective_bpm ≤ 120)
4. **Night (21:00 – 00:00)**: $(s_5 \geq 45 \vee s_4 \geq 50 \vee s_3 \geq 55) \wedge (\text{Arousal} \leq 0.15 \wedge \text{Valence} \leq 0.25) \wedge (\text{effective_bpm} \leq 115)$(s5 ≥ 45 ∨ s4 ≥ 50 ∨ s3 ≥ 55) ∧ (Arousal ≤ 0.15 ∧ Valence ≤ 0.25) ∧ (effective_bpm ≤ 115)
5. **Midnight (00:00 – 05:00)**: $(s_1 \geq 55 \vee s_3 \geq 55) \wedge (\text{Arousal} \leq 0.0) \wedge (\text{effective_bpm} \leq 90)$(s1 ≥ 55 ∨ s3 ≥ 55) ∧ (Arousal ≤ 0.0) ∧ (effective_bpm ≤ 90)

Circadian Fallback: Guarantees every track matches at least 1 Time-of-Day channel based on its closest valence/arousal quadrant.

Special / Global Channel (1 Channel)

- **Shuffle All Songs**: Always assigned to all analyzed tracks.

6. Sony SMFM Binary Block Layout (3,270 Bytes)

Every chunk follows a 20-byte envelope header: `[4B Tag][4B Magic "STAE"][4B Magic "MMLW"][4B Flags 0x01008000][4B Length (Big Endian UInt32)]`

```
Offset      Size    Tag     Description
--------------------------------------------------------------------------------
0x0000      4       SMFM    Global Application Header Magic (ASCII "SMFM")
0x0004      20+4    GBPM    Global BPM Chunk (IEEE 754 Big-Endian 32-bit Float)
0x001C      20+36   STBF    Sony Track Beat Features (Rhythmic Complexity Vector)
0x0054      20+8    STSA    Sony Track Sabi / Chorus Highlight [Start_ms, End_ms]
0x0070      20+160  STMO    Sony Track Mood Scores (40 entries x 4 bytes: [0x01, Mood, Q, Score])
0x0124      20+160  STHF    Sony Track Harmonic Features (40 entries x 4 bytes: [0x08, Bin, Val, 0x00])
0x01D8      20+2778 STMM    Sony Track Measure/Melody Map (Embedded GBPM floats at +32, +90)
--------------------------------------------------------------------------------
Total Size: Exactly 3,270 Bytes
```

Storage Encapsulation by Format:

- **FLAC (`.flac`)**: Encapsulated in FLAC `APPLICATION` metadata block (block type 2) with application ID `SMFM` (0x53 0x4D 0x46 0x4D).
- **MP3 (`.mp3`)**: Encapsulated in ID3v2.3 `GEOB` frame (description `"SMFM"`) and `PRIV` frame (owner `"SMFM"`).
- **M4A (`.m4a`)**: Encapsulated in MP4 freeform metadata atom `----` with mean `com.sony.walkman` and name `SMFM`.

7. Comparison & Review Checklist for User

Please review the following items against your official Sony 12 Tone SensMe documentation:

1. **Tempo Half-Speed Convention**:
    - In Sony Walkman firmware, tracks with $\text{BPM} < 58.0$BPM < 58.0 are often doubled ($2 \times \text{raw}$2 × raw), while in some Sony tools, tempos > 75> 75 BPM are stored as half-tempo (e.g. 84 BPM stored as 42.0f).
    - *Question for comparison*: Does your Sony documentation specify whether GBPM in `GBPM` chunk stores actual BPM or half-BPM?
2. **STMO Mood Slot Ordering**:
    - Walkman Studio uses: `[0: Energetic, 1: Relax, 2: Upbeat, 3: Mellow, 4: Emotional, 5: Lounge, 6: Dance, 7: Extreme, 8: Acoustic, 9: Electronic]`.
    - *Question for comparison*: Does the slot indexing match Sony's internal array order?
3. **Channel Activation Thresholds**:
    - Currently, mood scores $\geq 50 \ldots 55$≥ 50 … 55 trigger channel qualification.
    - *Question for comparison*: Does Sony define fixed integer score cutoffs, or a percentile rank among all songs in the library?
4. **Circadian Time-of-Day Partitions**:
    - Walkman Studio segments the day into:
        - Morning (05:00 – 10:00)
        - Daytime (10:00 – 17:00)
        - Evening (17:00 – 21:00)
        - Night (21:00 – 00:00)
        - Midnight (00:00 – 05:00)
    - *Question for comparison*: Are these hour boundaries and mood mappings identical to Sony's official firmware clock?

Verification Plan

Automated Tests

Run the dedicated SensMe DSP and binary round-trip test suite:

```
bash
./venv/bin/pytest tests/test_senseme.py -v
```

Expected output:

- `test_schema_constants` PASSED
- `test_payload_builder_exact_match` PASSED (validates exact 3,270-byte match against hardware Walkman file)
- `test_walkman_14_channels` PASSED

Next Steps Once User Provides Adjustments

1. Fine-tune DSP feature weights or threshold constants in `backend/senseme_dsp.py` and `Sources/WalkmanStudio/SenseMeEngine.swift`.
2. Run test suite to verify binary payload synthesis.
3. Recompile release app (`./build_app.sh arm64`) and restart.